

Brazilian E-Commerce by Olist

Exploratory Data Analysis

Name: Ahmad Assi

Problem Statement:

Olist is an e-commerce platform that connects small and medium-sized businesses to customers. The platform operates as a marketplace, where merchants can list their products and services and customers can browse and purchase them online.

As a new data analyst at Olist Team, you have been tasked with analyzing the most in-demand products, identifying the most active sellers and their impact on overall sales, and examining delivery performance to determine its relationship with low ratings. The goal of this analysis is to provide comprehensive insights to boost sales and enhance the customer experience.

Business Impact:

This analysis provides actionable insights to enhance sales and customer experience:

- **Optimizing inventory and marketing** by identifying high-demand products. For example, this helps ensure that top-selling items are always in stock and allows for more effective, targeted marketing campaigns.
- **Strengthening relationships with top sellers** by recognizing their contributions to the platform and offering incentives to encourage continued high performance and reliability.
- **Improving delivery performance** by identifying and addressing delays that negatively impact customer reviews. Timely deliveries are key to ensuring a positive customer experience and encouraging repeat purchases..
- **Enhancing pricing and promotion strategies** by leveraging insights on customer behavior and sales trends.

By implementing these findings, Olist can drive higher sales, improve decision-making, and enhance overall platform efficiency.

General Dataset Information:

File Name:	archive
Description:	Orders at Olist Store from 2016 to 2018
Dataset Details:	100k Rows & (52 Columns at 9 files csv)
Size:	42.6 MB (44,717,580 bytes)
Source:	Kaggle - Dataset Link

Target Features:

The columns that will be used to simplify and analyze the problem statement are:

1. To identify the most in-demand products

- **order_id** : order id per product
- **product_id** : product id
- **product_category_name_english** : Product category name in English
- **price** : product price

2. To identify the most active sellers and their impact on sales

- **seller_id**: seller id
- **seller_city, seller_state**: Seller's location
- **order_id** : order id per seller

3. To analyze the relationship between Delivery Days and Review Score

- **order_purchase_timestamp**: order purchase timestamp
- **order_delivered_customer_date** : order delivered customer date
- **DeliveryDays**: A feature engineered column for Number of delivery days
- **Review_Score**: Ratings
- **customer_city, customer_state**: customer's location

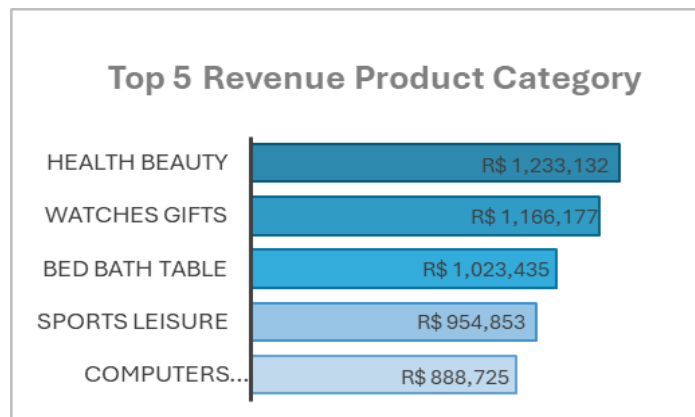
• List of Analysis

Analysis #1 - the most requested products

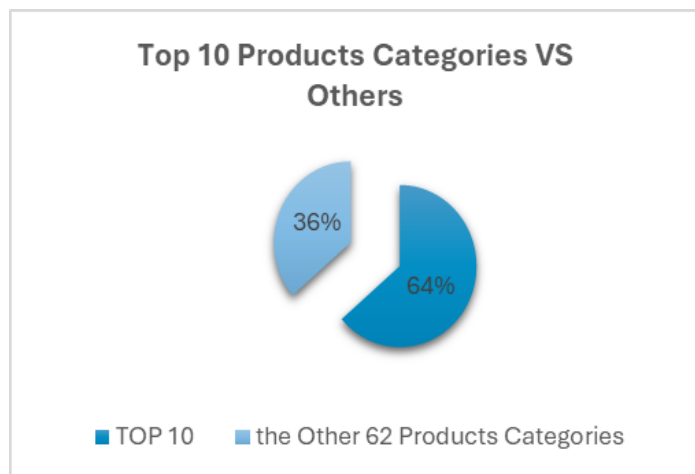
- total number of orders for each product was calculated using (Pivot Table).

Product Category	Total Revenue
computers accessories	R\$ 888,725
sports leisure	R\$ 954,853
bed bath table	R\$ 1,023,435
watches gifts	R\$ 1,166,177
health beauty	R\$ 1,233,132

- Top 5 Products Categories Revenue (**Bar Chart**): “Health Beauty” is the top-performing product category, suggesting a strong customer preference in this sector.



- Using (**Pivot Table** and **Pie Chart**) 64% of Orders are concentrated only in 14% of products Categories, which indicates the concentration of sales on a few popular products.
- The remaining products have less demand, which requires improving marketing strategies for these products.



Analysis #2- States of the most active sellers and their impact on Sales

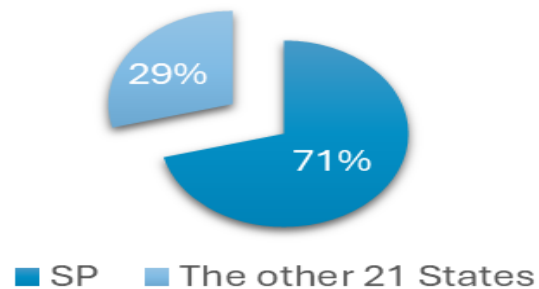
Sellers State	NumberSales_PerSeller
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- The total number of orders for sellers state was calculated by using (**Pivot Table** and **PieChart**).

SP	82,368
MG	9,060
PR	8,848
RJ	4,892
SC	4,188

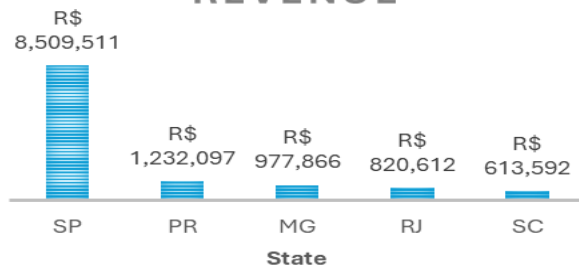
- 71%** of Orders are concentrated only in SP State , which indicates weak sales in other states.
- The Other States have less demand, which requires improving marketing strategies for these States.

SP State Vs Other States Sales



- Top 5 Seller State Revenue (Bar Chart):** There's a massive gap in revenue between states, with São Paulo (SP) generating 64% of the revenue.

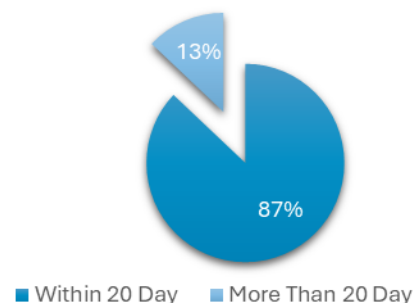
TOP 5 SELLER STATE REVENUE



Analysis #3- delivery performance and its relationship to Review Score

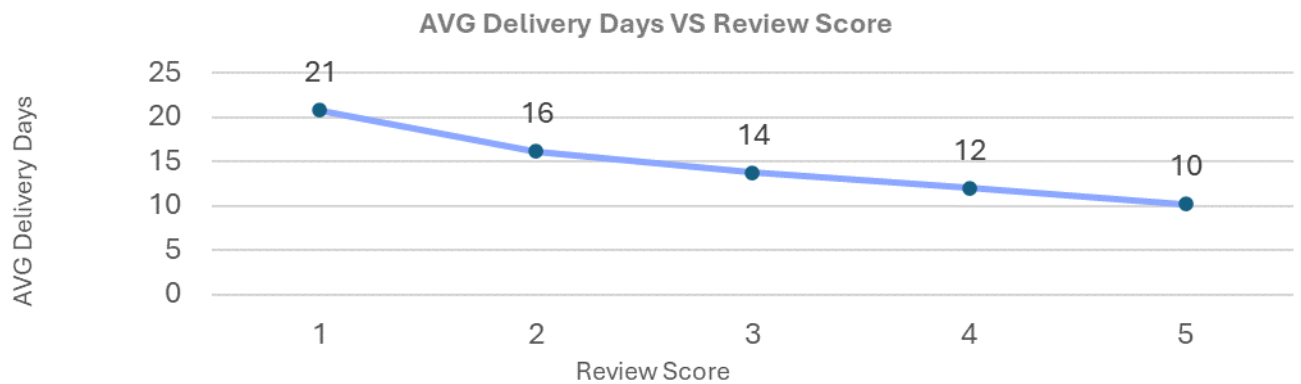
- Orders are classified according to delivery time using (**Pivot Table** and **PieChart**).
- 87% of orders were delivered Within 20 days, and 13% exceeded this period, which indicates that delivery performance is good

Order Delivery Performance



- **The relationship between Delivery Days and Review Score**

- There is a negative correlation between the number of Delivery Days and Review Score, meaning that faster deliveries tend to receive higher review scores the correlation coefficient (CORREL()= -0.3)
- The Line Chart shows the relationship between the Average of Delivery Days and Review Score, lower delivery days leads to an increase in the review score



Analysis #4- Basic Statistics

- Descriptive statistics were used to understand the distribution of prices, ratings, and number of orders.

measure	price	review score	# of orders by state
mean	124	4	39
median	79	4	8
St. Div	188	1	126
upper bound	286.75	5	63
lower bound	-107.25	1	-33
skewness	7.8990	-1.4580	9.6955

● Conclusion

- Our analysis of (**96,478 Orders**) across (**72 product categories**) reveals key insights regarding product demand, seller activity based on location, and delivery performance:
- **Most In-Demand Products:** 64% of Orders are concentrated only in 14% (TOP 10) of products Categories, with (**bed bath table**) leading in terms of orders.
- **Sales Distribution by Seller States:** The results indicate that (**SP State**) contribute the most to total sales for (**71%**), suggesting that the geographic distribution of sellers plays a role in demand and revenue. Improving less-selling products through targeted marketing strategies.
- **Delivery Performance & Review Score:** (**87%**) of orders are delivered within **20 days**, while (**13%**) experience delays exceeding **20 days**. Lower delivery days lead to increased Review score and thus customer satisfaction.
- **Market Performance:** Most products achieve (**38%**) of their Revenue potential, while only (**62%**) exceed (**\$8,600,000**) in sales. This indicates an opportunity for sellers to enhance pricing strategies, promotional efforts, and product visibility to increase sales.